

Abstract

Nano-paper is a biomaterial made of nano-fibrillated cellulose (NFC) in a similar method to that used in the creation of office paper. This project examined the difference in mechanical properties, primarily tensile strength, of two different types. Generation one was enzymatically pre-treated, whereas generation two was Carboxy-methylated (see Introduction for definitions). Three rounds of testing were carried out on each, using an Instron 1121 universal testing machine, to determine the tensile strength of each. Images of the sample edges were taken using a Renishaw Ramen Light Microscope. These were used to analyse different preparation methods and to highlight major structural differences between the samples.

The results of the testing showed that there were several flaws with the testing procedure. This centred on the cutting of the samples during preparation and the alignment in the testing machine. These lead to the results being very spread, and lower results than could have been measured. The testing did show though that the generation one samples had a higher specific strength on average, compared with generation one. Generation one had an average specific strength of 103.1 KNm/kg, with a maximum of 170.2 KNm/kg. Generation two had an average specific strength of 68.5 KNm/kg and a maximum of 189.4 KNm/kg. Several possible methods of improving the testing procedure, to produce more consistent and accurate results are included within the paper.

Keywords: Nano paper,

Nano-fibrillated cellulose,

Mechanical properties,

Tensile testing,

Specific strength

Table of contents

1. Introduction and background.....	1
2. Literature review.....	2
2.1. Understanding Nano-paper	2
2.2. Testing procedure.....	3
3. Methodology and theory.....	3
3.1. Aims of experimentation.....	3
3.2. Approach to testing	4
3.3. Sample Imaging.....	5
3.4. Calculations and analysis of results	5
3.4.1. Specific Strength.....	5
3.4.2. Images of edge defects	6
4. Experimental work	6
4.1. Sample Preparation	6
4.2. Tensile Testing	8
4.3. Sample Imaging.....	9
4.4. Issues identified regarding preparation and testing procedures	9
4.4.1. Sample Preparation.....	9
4.4.2. Testing Procedure	10
5. Presentation of experimental results and initial discussion	11
5.1. Generation One Samples.....	12
5.2. Generation Two Samples	14
5.3. Microscope Images	16
5.3.1. Generation one images	16
5.3.2. Generation two images	18

5.3.3. Overall results of imaging	19
5.4. Conclusion.....	20
6. Conclusion	22
6.1. Testing procedure	22
6.2. Comparing the two generations of nano-paper	23
7. Project management, consideration of sustainability and health and safety	24
7.1. Project management	24
7.2. Health and safety	27
7.3. Sustainability.....	28

1. Introduction and background

Nano-paper is a developing product in the field of renewable, sustainable and biodegradable materials. It is made of the same basic material as standard paper, cellulose. Cellulose is found in the walls of plant cells, making it cheap and abundant, and provides structure and strength to the plant. It is a polysaccharide consisting of long chains of glucose ($C_6H_{10}O_5$) molecules. These chains are connected via hydrogen bonds between the chains, which provide strength.

Cellulose materials have been used for centuries all over the world due to their availability and strength. Wood is around 35-40% cellulose and has been used to build homes, ships and furniture since recorded history. Cotton used for clothing is around 90% cellulose. NFC, which forms nano-paper, has the potential to form many useful composites, and on its own has the potential for technological and bio medicinal applications.

Nano-paper's important potential uses are based on its mechanical properties. Compared with standard office paper, nano-paper has a significantly higher tensile strength. This is due to the difference in the preparation methods used, as both are made from the same basic material. In both cases plant material is taken and cut up into mulch. The difference is that with standard paper the fibres are normally in the range of X , with nano-paper the fibres are significantly smaller. Through the use of chemical or enzyme treatment it is possible to break down the cell walls and produce fibres in the nanometre range. These small fibres are known as nano-fibrillated cellulose (NFC). The reduction in fibre size dramatically increases the total surface area to volume ratio of each fibre, leading to more hydrogen bonds within any given volume of paper, and so increasing the strength of the material. The mulch is then dried as sheets in the same way as for standard paper, though normally in much smaller batches.

The aim of this project is to investigate the mechanical properties of the nano-paper. Two nano-paper samples were provided from Sweden. The first, generation one, has been enzymatically pre-treated. This means that enzymes were used to break down the cell walls. The second, generation 2, was treated with Carboxy-methylate, which breaks down the cell walls chemically.

The main metric was the tensile strength of the samples. The results were then compared to values from other papers where similar tests were carried out to assess their likely accuracy. They were also compared to other materials to assess how nano-paper compares to materials that are currently widely used. The strain at failure was examined for the samples. It was also important to assess the testing methods to ensure that the most accurate results possible were achieved.

2. Literature review

2.1. Understanding Nano-paper

There are several papers looking at nano-paper, both its applications and the way it is made. These helped provide a background understanding of the material and a better understanding of the reasons for its high specific strength, which assisted in the writing of section 1. S.J.Eichhorn et al. (2010)ⁱ was used to research the structure of nano-paper and the methods used in its production. This gave an overview of the different methods of fibre preparation and how sizes affected strength.

Dieter Klemm (2011)ⁱⁱ, as well as giving an overview of the production process, discussed the possible applications of the material. These included as security papers, food packaging, and combining with resins to produce high strength composites. Medical applications are also discussed, such as wound dressings. The paper also discussed the fact that the material is biodegradable, and readily degrades in the presence of water and other substances. This influenced the conclusion that the adhesive used during testing could be degrading the samples, as mentioned in section 4.4.1.

Guangyuan Zeng et al. (2013)ⁱⁱⁱ not only discusses what cellulose is, but also many of the new applications for cellulose in the energy industry. The paper can be used for energy storage systems that work by using liquid electrolytes. Further as the paper has a hydrophilicity nature it can be used in types of electrodes. The flexibility of the paper is one

of the main advantages, and research is also being carried out into its use for ultracapacitors and in LI-ion batteries. For these applications strength is important, as well as the flexibility and transparency.

Marielle Henriksson (2008)^{iv} gave expected values for the strength of cellulose nanopaper. During tensile testing a strength of 214 MN/m² were observed. While this will be influenced by the density of the sample, as is stated in the paper, it gives an idea of the expected strength. Based on the data given in the paper for density, 1080 kg/m³, a specific strength of 198 KNm/kg can be calculated using the equations from section 3.4.1. This again can be compared against the measured values.

2.2. *Testing procedure*

There are few high level scientific papers that discuss the tensile testing process in detail. Therefore it was necessary to use papers that weren't specifically about cellulose nano-paper. Fortunately though the testing process is similar for all samples of similar dimensions. ASM International (2004)^v provided a comprehensive overview of the testing procedure. This helped when issues arose, as it made easier to understand what may be causing them. Dr Tommy Shyng also provided advice on the testing procedure.

Due to the lack of detailed information on issues surrounding the tensile testing procedure for nano-paper part of this report will look to investigate issues that arise. Different techniques of preparation and testing will be investigated and discussed to observe the effect. This will show the main issues that are caused during the testing process. It potentially could also lead to improvements in the testing procedure.

3. Methodology and theory

3.1. *Aims of experimentation*

The aim of this investigation was to analytically determine the tensile strength of the two different NFC samples. This was the principle metric used to compare the papers produced

using the two different methods of NFC preparation. To calculate this the density and the breaking force for the sample needs to be known. The density is calculated by measuring the volume of the sample; multiplying width by length and thickness, and dividing the mass of the sample by this. The breaking force is the maximum tensile force that the specimen can withstand before failure occurs. This will be tested using an Instron 1121 universal testing machine.

3.2. *Approach to testing*

Due to the limited available number of samples that could be prepared it was necessary to ensure that there would be enough samples for meaningful conclusions to be drawn from each round of testing. Therefore it was initially decided that two rounds of testing would be carried out to allow any improvements to the procedure to be made after the first round. However, after the first round of testing, it was decided that it would be beneficial to add a third round of testing, to remove a factor which was believed to have impacted the results of the first round of testing (see below Section 4.4). Cut offs were used for the extra round to ensure sufficient numbers of samples were available.

The initial round of testing was carried out with approximately half the samples. This was partially used to assess the procedure by highlighting any issues and ensuring the correct load cell was used on the testing machine. The samples were prepared in the standard way for this type of testing, as described in section 4.1.

The methodology of the secondary round was adapted to address the issues highlighted in the preliminary round, as discussed in section 4.1. This was done primarily to assess the effect of two different methods used to cut the samples, and to assess the effect that using an adhesive on the sample had had on the results of round one. Images of the edges of the samples were taken at this stage using a light microscope as a qualitative method of evaluating the effect of the cutting methods. A third round of testing was then undertaken using fresh samples to implement changes to remove the impacts observed on the round 1 samples from the cutting methods.

3.3. Sample Imaging

Results from the first round of tensile testing were dramatically lower than expected for both samples. In the case of the Generation 2 samples (chemically treated,) this was believed to be due to the cutting method during preparation. Initially a Stanley Knife was used, but this often led to tearing of the sample edge. This is an issue as edge defects will dramatically reduce the force needed to break the sample, due to providing an area of stress concentration (Dr.ir. P.J.G. Schreurs, 2012)^{vi}. Therefore a new cutting method needed to be identified which would reduce the degree of edge defects.

Biological scissors were selected, as they are sharp enough to cut through the sample cleanly and so would pull less on the sample as they cut. To back up numerical results, images were also be taken so that the different preparation methods could be examined. The images produced could then be compared by eye to assess the difference in magnitude and number of edge defects. These were then used to decide what preparation method would be used for the third and final round of testing. The testing method is discussed in section 4.3 and the images are included in section 5.

3.4. Calculations and analysis of results

3.4.1. Specific Strength

Initially it was necessary to calculate the strength of the sample. This was done using equation 1 (M.F. Ashby, D.R.H. Jones, 2012)^{vii};

$$\text{Strength} = L/A$$

Equation 1; Strength formula, measured in N/m², L is breaking load in N, and A is the sample cross sectional area in m²

The density of the sample was then calculated using the formula shown in equation 2;

$$\rho=m/V$$

Equation 2; Density formula, ρ is density measured in kg/m³, m is mass measure in kg, and V is volume measured in m³

The values for this were measured according to the techniques described in section 4.1. These were then used to calculate the average density for each of the generations of nano-paper. The average density and strength were then used to calculate the specific strength using equation 3;

$$\text{Specific strength} = \text{strength} / \rho = (L/A)/(m/V)$$

Equation 3; Specific strength equation, measured in Nm/kg, L is load at failure given in N, A is the cross sectional area given in m², m is mass measure in kg, and V is volume measured in m³

3.4.2. Images of edge defects

The results were analysed in a purely qualitative way. The images were analysed by eye as it was felt the difference would be dramatic enough that this would be sufficient. This was particularly true in the case of generation 2 samples where the preparation technique was felt to have the biggest effect on recorded tensile strength.

4. Experimental work

4.1. Sample Preparation

Samples were prepared according to instruction from Dr Tommy Shyng, Exeter University, as well as information from scientific papers, section 2.2. The initial method of sample preparation is the standard for this type of testing, based on the papers mentioned in section 2.2. The methods of preparation changed after the initial round of testing due to issues that were highlighted. Therefore the preparation methods for the second and final rounds were different.

As the NFC samples came in the form of several small circular sheets, there was an obvious limit to the testing strip size. Due to the number of sheets provided a width of 10mm and a

gauge length of 35mm was chosen, allowing 2 samples to be cut from each sheet. It was decided that this would be a sufficient number of samples to draw a conclusion from. These were then stuck, using glue, to a card window to make the sample more rigid and to protect it from the grips of the testing machine. The glue used was “Gorilla Super Glue”, a modified cyanoacrylate. A labelled diagram of a sample is shown in Figure 1 All the samples for this round of testing were cut using a Stanley Knife.

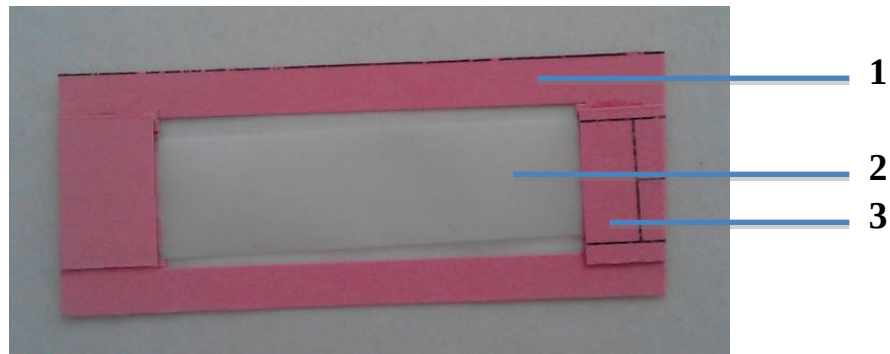


Figure 1; Diagram of a sample prepared using a card window, 1: card frame, 2: nano-paper sample, 3: card piece used to protect the sample

The second round of testing was carried out to address issues identified with the results from the initial round. These issues were; the potential impacts of the adhesive and the cutting methods on the results, which were considered as having potentially impacted the results of the round one tests. For the second round no glue was used and the effect of cutting the NFC using surgical scissors was investigated. The reasons for this are explained in section X. Due to the importance of maintaining a sufficient amount of NFC sheets, to allow a sufficient number of samples, cut offs from sheets used in round one were used in round two. Again these were cut to a rectangular shape, but the size varied, with the aim being to gain the largest possible strip from each piece. These were not glued to a window this time. Instead small pieces of card were placed either side of the strip in the jaw to protect the sample. This meant that any possible effects on the tensile strength of the samples in the first round of tests from the chemicals within the glue, would be removed from the second round tests. All generation 2 samples were prepared using the scissors. Samples of generation 1 were prepared using both the scissors and the Stanley Knife, to allow for comparison, to highlight any impact on tensile strength from the two different methods of cutting.

For the third round of testing the remainder of the NFC sheets were used. For this round the aim was to produce the longest strip possible whilst achieving 3 strips per sheet, due to there

being less available sheets. As for round two, no glue was used on the samples. This meant that the gauge lengths for these samples would be determined at the testing stage. This was unlike the first round, where it was determined by the size of the card window. For this the third round, generation one samples were prepared using the Stanley Knife and generation two using the surgical scissors. The reasoning for this is discussed in section X.

After all samples were prepared they were then weighed using a digital balance accurate to 0.0001g. This was then repeated for the entire sample after the test to measure any losses occurring during the testing process. The sample was then cut so that the tested area could be weighed. The width and thickness of the samples was measured using a ruler and a micrometre respectively. The gauge lengths for the first round of testing came from the length of the window hole, which was measured using a ruler. For the second and third tests the gauge length was measured using a ruler once the sample was within the testing machine.

4.2. Tensile Testing

The tensile testing was carried out using an Instron 1121 universal testing machine. This was fitted with a 500N load cell and set at a crosshead speed of 1mm per minute. The samples were then loaded into the jaws as shown in figure 2 and the machine was set to run. A strain of 15% was chosen as an end point as this was significantly beyond the expected strain at failure.

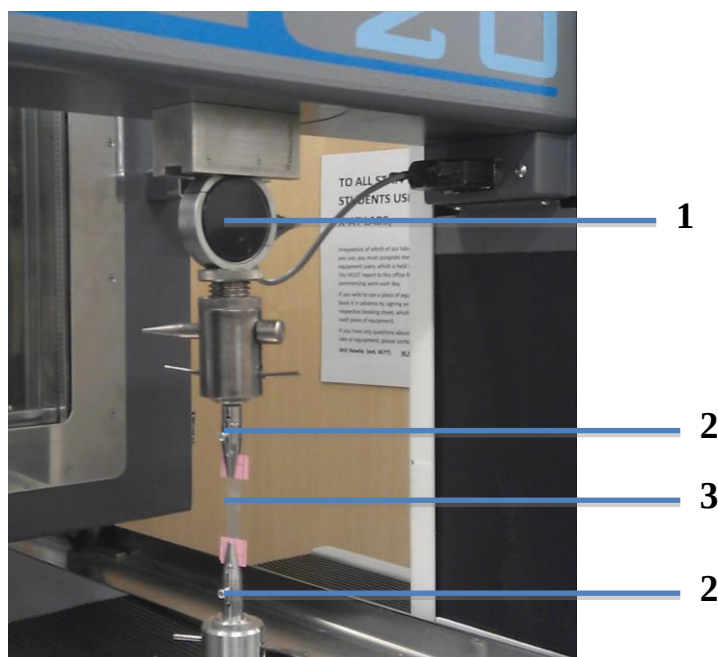


Figure 4; Diagram of the tensile testing machine, 1: load cell, 2: clamps, 3: sample

4.3. *Sample Imaging*

Two samples for each of the two types of NFC were prepared for imaging. For each type of NFC one sample was cut using biological scissors, and the other using a Stanley Knife. These were then loaded onto glass slides and images were taken using the Renishaw Ramen Light Microscope. These images were taken of the edges of the samples at both 10x and 40x magnification, with the aim of showing the different levels of edge defects caused by the different methods of sample preparation.

4.4. *Issues identified regarding preparation and testing procedures*

4.4.1. *Sample Preparation*

There were two main issues with the preparation techniques that were believed to lead to the lower than expected results for the breaking force. The first was the use of glue to secure the strips of NFC to the card windows. The second was the method of cutting the strips for testing. In both cases changes were made with the aim of verifying whether the issue did, in fact, affect the tensile results of the first round of tests and to produce results from the second and third stage of testing which were isolated from these factors.

The first issue was the use of adhesive to attach the strip to the card support window. During the first round of testing all of the generation one samples, the ones treated enzymatically, broke at the point where they connected to the window. This was an issue as it indicated that the point of contact with the card window was consistently the weakest part of the paper. This suggested that the results potentially indicated a lower tensile strength than was the case for the rest of the sample.

However, this was not the case for the generation 2 samples. This could be due to the fact that edge defects were causing the samples to fail at a much lower force. Another reason could have been that, whilst the enzymatically treated samples were affected by the chemicals in the adhesive, the chemically treated samples were chemically different enough not to be affected.

It was concluded that, where this effect was observed, it was due to a chemical within the adhesive degrading the sample in the time between preparation and testing. This effect could have produced results which indicated a lower tensile strength than would otherwise be the case by producing weak points within the sample which would not otherwise exist. Therefore it was decided that for subsequent rounds of testing no adhesive would be used. This meant that the no support window could be used. Instead, 2 card squares were used at either end of the strip to prevent the sample being damaged by the jaws of the testing machine.

The secondary issue was the presence of edge defects. Whereas the use of a Stanley to cut generation one samples produced a flat looking edge, it was difficult to achieve any level of smoothness for generation two. This is believed to have lead to the dramatically lower results for breaking load for generation two samples. It was important to remove this potential impact and produce results which would not be impacted by the less smooth edges of the generation 2 samples. Biological scissors were chosen as a potential substitute for the Stanley Knife. It was hoped that they would not pull as much on the sample and were sharp enough to produce a clean cut. Due to both qualitative and quantitative results indicating an impact on the generation 2 samples, it was decided to use the Stanley knife for generation one and the biological scissors for generation two. The reasoning behind this is explained further in section 5.

4.4.2. Testing Procedure

The main process control with the testing procedure was ensuring that the sample was straight between the two jaws of the testing machine and equally taut on both sides. This was more easily achieved for the first round of testing, where the presence of the window provided a level of structure,. However, for the later rounds significant care was necessary to ensure consistent loading into the jaws to produce a fair test. This was important because, if the sample was equally taut on both sides then the force applied would differ through out the sample. This would mean that the measured force would be concentrated on a particular point within the sample and so the sample would fail under a lower load than would be the case if the tensile stress were applied evenly throughout the material, producing results which were

artificially low.

Secondly slipping was much more common with the second and third rounds of testing. This was concluded to be due to the sample not being secured to the window, and instead cardboard squares being used. The static friction between the card and the sample was possibly lower than that between the card and the jaws. This would lead to the sample slipping. The easiest way to prevent this was to angle the jaws so they gripped the sample with a smaller surface area, therefore exerting a larger pressure. However, this had the potential to affect the results for the sample due to uneven application of load (as referred to above), so it was necessary to ensure that this was carefully balanced.

5. Presentation of experimental results and initial discussion

The raw results have not been included as it was felt that nothing was gained from their inclusion. Instead, summaries of the results are included in the tables present in the following sections. For the results in these sections certain shorthand notation is used;

- G1 represents generation 1 samples
- G2 represents generation 2 samples

These results will be used to assess the testing procedure and to calculate the specific strength of each generation of nano-paper. These will then be used to compare the two types of nano-paper. The results will also allow the cellulose nano-paper to be compared to other materials.

Due to small sample sizes differences between results need to be significant to be meaningful. Because of this, standard deviations have been included in the tables so that a margin of error can be approximated. Differences of less than 1 standard deviation will not be deemed significant for the purposes of this report.

5.1. Generation One Samples

The table below, Figure 3, contains the summarised data for the three rounds of testing of generation 1 samples. The first graph shows the spread of breaking loads for the generation 1 samples, Figure 4. The second graph, Figure 5, displays the values for specific strength for the generation 1 samples. Because specific strength takes into account the size and density of the sample samples of different sizes can be compared.

	Strength / MN/m ²			Specific Stength / KNm/kg		
	Maximum	Average	Standard Deviation	Maximum	Average	Standard Deviation
Round 1 Stanley Knife	272.9	215.0	58.5	170.2	134.0	36.5
Round 2 Scissors	162.0	114.5	32.3	98.6	69.7	19.7
Round 2 Stanley Knife	241.6	161.7	57.9	147.0	98.4	35.2
Round 3 Stanley Knife	230.1	182.6	38.0	125.8	99.9	20.8

Figure 3; Table summarising the results of testing for generation one samples

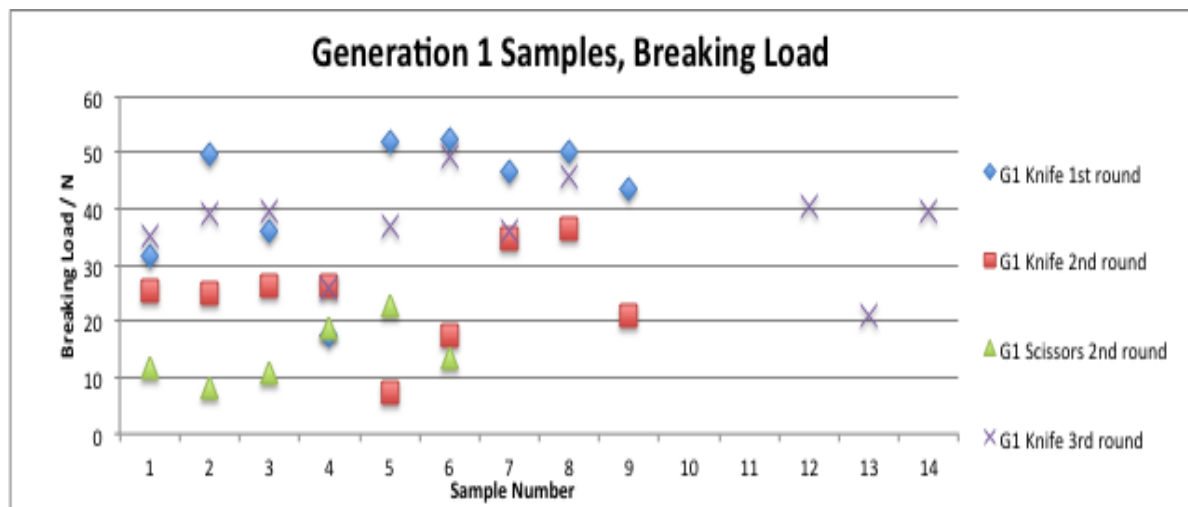


Figure 4; Graph showing the load at which failure occurred for generation one samples

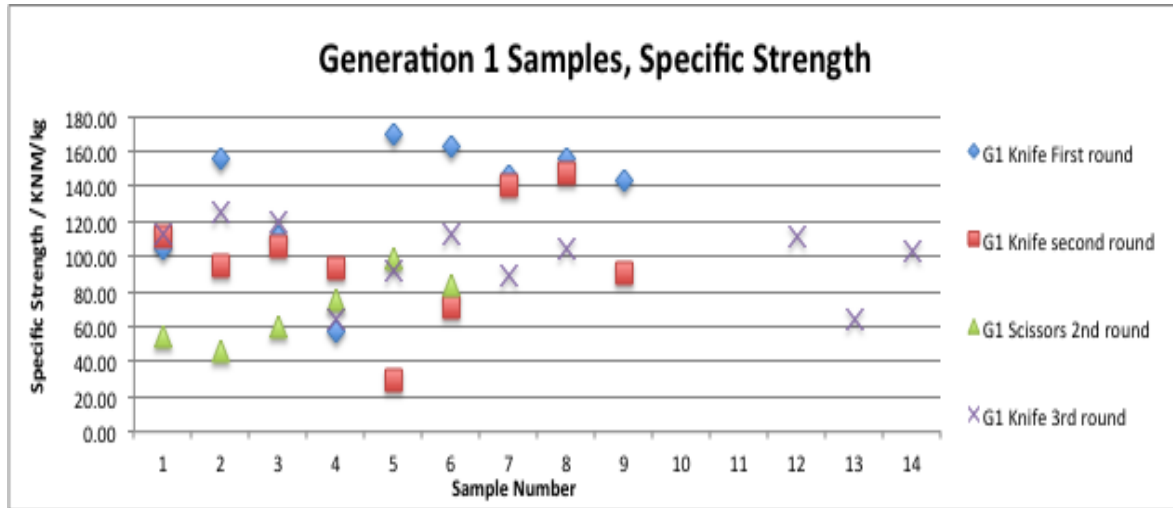


Figure 5; Graph showing specific strength for generation one samples

The results from rounds one and two of testing were used, along with the images discussed in section 5.3, to determine the best method of cutting the sample. The two options were using a Stanley knife or biological scissors. The results for scissors from round two show that the Stanley knife is the better choice for generation one samples. The table shows that the values for strength and specific strength are considerably lower for the samples cut with scissors, compared to those cut with the knife. This still holds up when the standard deviation is taken into account. There is significantly over 1 standard deviation difference between the averages from round one (using the Stanley knife) and round two (using the scissors). The results for the round two scissor samples are also lower than for the round two knife samples by greater than one standard deviation. These results support the use of the Stanley knife to prepare generation one samples.

The graphs and standard deviations show that there is a large spread of results, regardless of the preparation method. This is likely to be caused by deformation in the samples, particularly edge deformation. This reduces the breaking load of the sample significantly through the creation of high stress areas. This leads to the conclusion that the average specific strength calculated is lower than the true specific strength of the material. This is backed by the fact that during the first round of testing all of the samples broke at the jaw. This is believed to be due to the adhesive that was used reacting with the sample, as the number where this happened dropped during later tests, in which no glue was used. If the samples were weakened then the average strength recorded in the first round is likely to be lower than it should be. Therefore it would be fair to assume that the specific strength of the generation one nano-paper is in excess of 134.0 KNm/kg. The highest specific strength recorded was

170.2 KNm/Kg. As this is just within one standard deviation from the average, and the low average is likely to be due to deformations, it is probable that this value is a more accurate specific strength, and not an error.

The results show a significant drop in all values after the first round of testing. This could be due to the effect of not using the cardboard window for these later tests. Not using the window caused issues with loading the sample into the tensile tester, as discussed in section 4.4.2. There is also the possibility that the samples deteriorated over time. It was necessary to see if this pattern continued for generation two samples to assess whether this gives further indications as to the reason for this.

5.2. *Generation Two Samples*

The table below, Figure 6, contains the summarised data for the three rounds of testing of generation 2 samples. The graphs, Figure 7 and 8, then show the values for breaking load and specific strength for generation two samples respectively.

	Strength / MN/m²			Specific Strength / KNm/kg		
	Maximum	Average	Standard Deviation	Maximum	Average	Standard Deviation
Round 1 Stanley Knife	189.9	144.4	38.8	114.2	86.8	23.3
Round 2 Scissors	311.3	104.5	77.1	189.4	63.6	46.9
Round 3 Scissors	111.6	85.9	22.4	79.1	60.9	15.9

Figure 6; Table summarising the results of testing using generation two samples

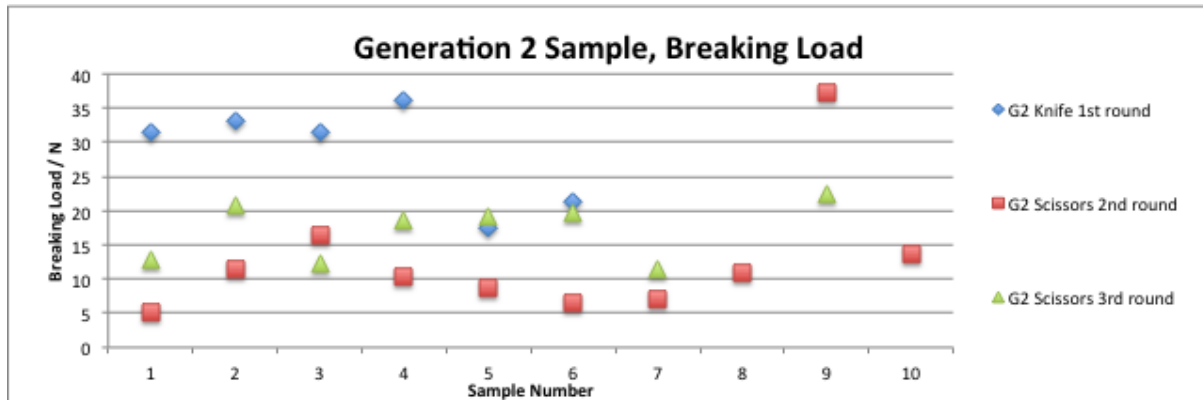


Figure 7; Graph showing the load at failure for generation two samples

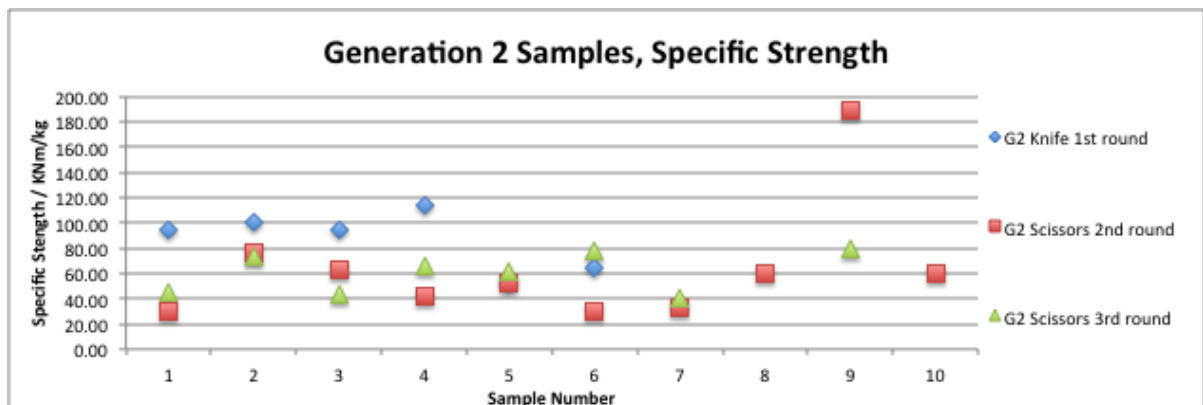


Figure 8; Graph showing the specific strength for generation two samples

The results from round one and two were used to help inform the choice of preparation method for the third round. The average results for these two rounds lead to a conclusion that the Stanley knife is the better tool for sample preparation. However, once standard deviations are taken into account, the difference becomes less significant, as neither is greater than one standard deviation apart. Further, the higher maximum is for the samples prepared using the scissors. The graph of specific strength shows that the results for the first round were higher though, and that an outlier skews the round two results. Because the results are not conclusive the microscope images helped to inform the decision to use scissors for round three.

Again the tables and standard deviations show a large spread, which is likely to be due to deformations within the samples. Further the results of round two and three are lower than those of round one, though not as significantly as for generation one samples. This lead to the conclusion that the results calculated for specific strength are lower than is accurate. This may be a due to a combination of deformations and issues with the preparation and testing

procedure. Therefore the specific strength of the generation two samples is considered likely to be greater than 86.8 KNm/kg. This suggests that the maximum value of 189.4 KNm/kg is likely to be due to a well prepared sample, with a low number of critical defects being introduced during cutting, rather than an error.

5.3. *Microscope Images*

Images of the samples were taken to be used, in conjunction with numerical data, to determine the preparation technique to be used for the final round of testing. For both generations of nano-paper a sample was prepared using a Stanley knife, and a second using biological scissors. These were taken using the Renishaw Ramen light microscope at both 10x and 40x magnification. It was discovered after the images were taken that they did not include the scale bars. Fortunately the magnifications of the images are known and the differences between samples should be large enough that conclusions can be easily drawn.

For all the images, the nanopaper is the section at the top of the image, and the section at the bottom is the glass slide.

5.3.1. *Generation one images*



Figure 9; Generation 1 Knife 10x

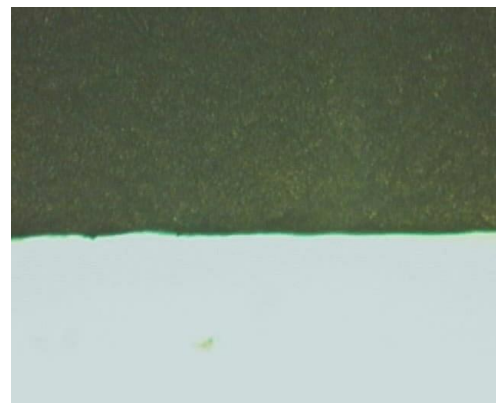


Figure 10; Generation 1 Scissors 10x

The two images above, Figure 9 and Figure 10, show the images taken at 10x magnification of generation one samples. At this magnification there appears to be very little difference in the amount of edge deformation caused by the two preparation methods. Because of this it was necessary to increase the magnification to 40x.

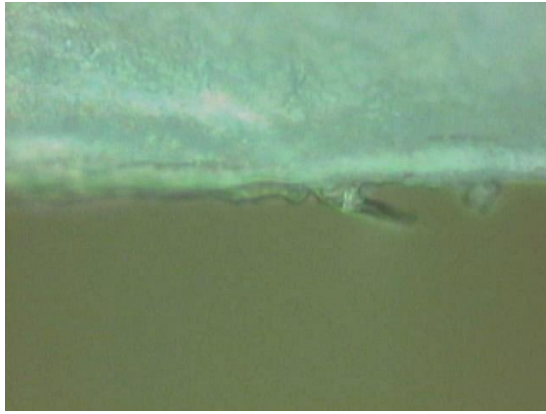


Figure 11: Generation 1 Knife 40x

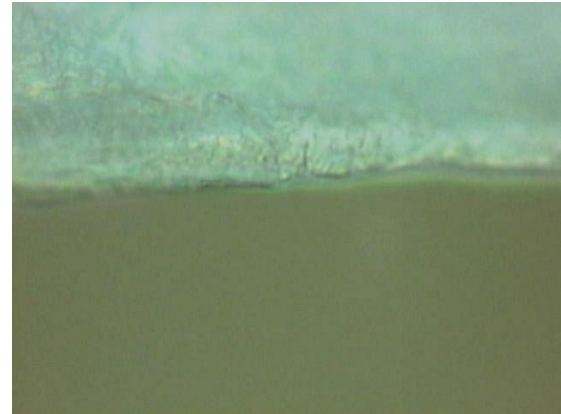


Figure 12: Generation 1 Knife 40x

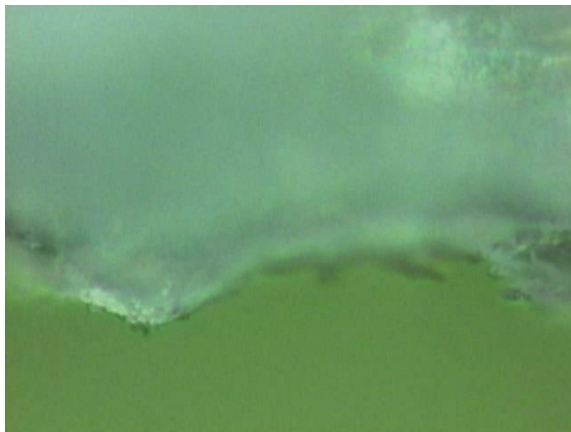


Figure 13: Generation 1 Scissors 40x

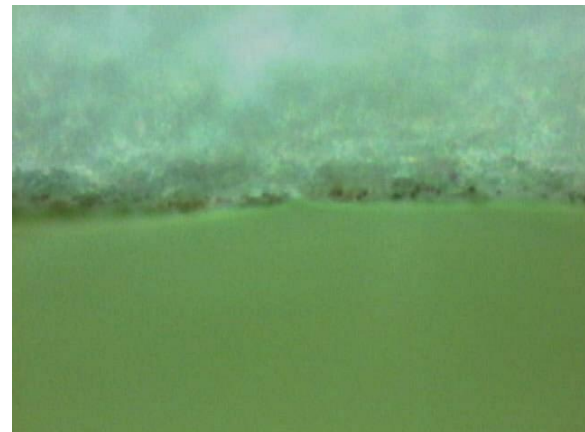


Figure 14: Generation 1 Scissors 40x

The lack of focus in the images is due to multiple layers being present, not the inability to focus of the operator. This is because there are sections that are in focus, and sections that aren't. This is caused by some sections being closer to the lens than others. These images show that whilst there are minor edge deformations for the samples cut using a knife, they are smaller than those present in the sample cut using scissors. The 'blurriness' shows that cutting the sample with scissors not only leads to the edge not being straight, but also multiple layers being formed. From this data it appears that the best cutting method for the samples is to use a Stanley knife instead of biological scissors. It is important to note though that all the images show some level of deformation at the edge of the samples, and these

could lead to lower results for strength values than would be expected or accurately reflect the strength of the body of the paper.

5.3.2. Generation two images

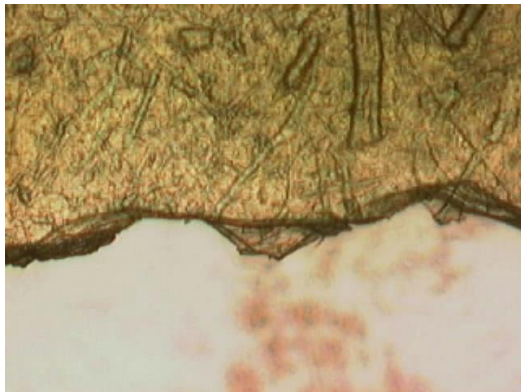


Figure 15; Generation 2 Knife 10x

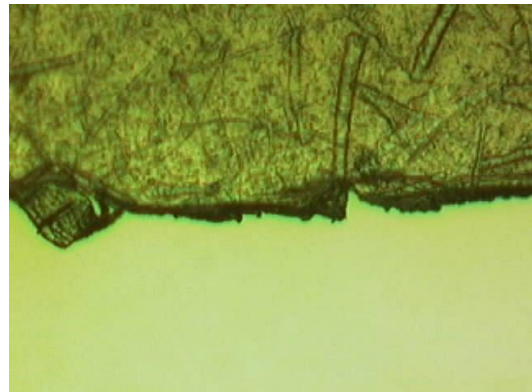


Figure 16; Generation 2 Knife 10x



Figure 17; Generation 2 Scissors 10x

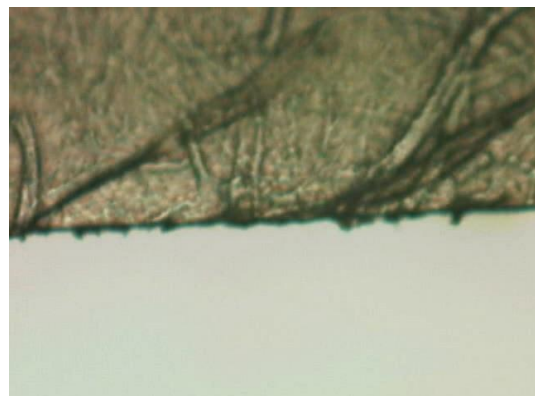


Figure 18; Generation 2 Scissors 10x

The above images are used to compare the effects of the two preparation methods on the generation two samples. These images show a large difference between the samples cut using the Stanley knife and the biological scissors, even at the 10x magnification. The scissors appear to create a significantly smoother edge with less defects. It is also possible to see the creation of multiple layers created at the edge where the Stanley knife has been used. For comparison with the generation one samples images were also taken at 40x magnification.

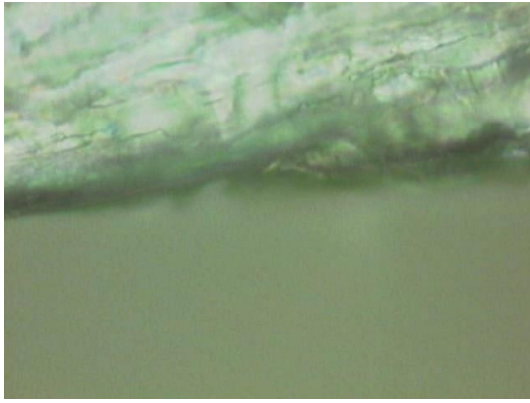


Figure 19; Generation 2 Scissors 40x

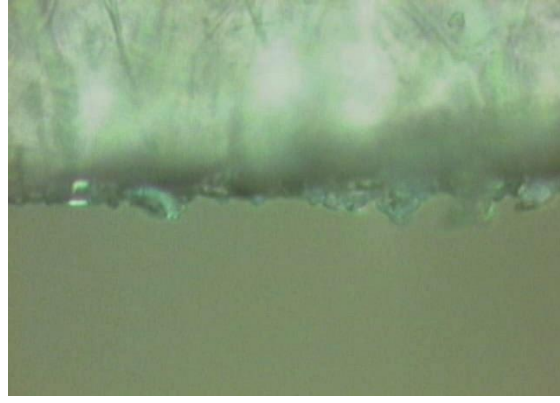


Figure 20; Generation 2 Scissors 40x

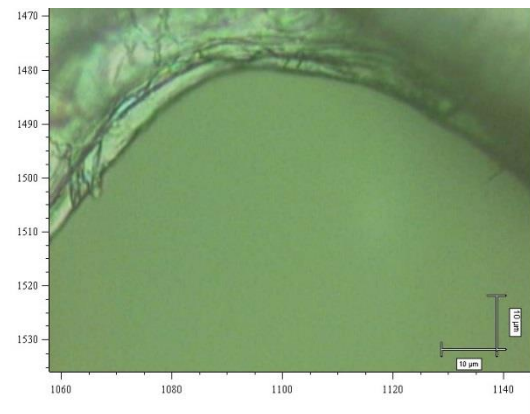


Figure 21; Generation 2 Knife 40x

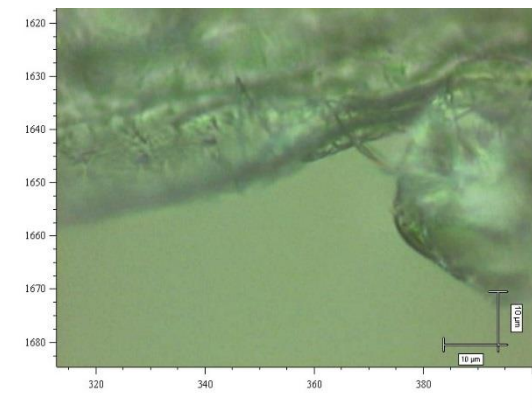


Figure 22; Generation 2 Knife 40x

The samples prepared using the Stanley knife show large deformations, confirming the conclusions drawn from the 10x images. The samples cut with the biological scissors are still significantly better than those cut with the knife, but edge defects are visible. These are shown by both the partial lack of focus of the images and the lack of smoothness of the edge. This will lead to significant reduction in the measured specific strength of the samples. However biological scissors are a significant improvement on the Stanley knife for preparing generation two samples. This was the reason for the use of biological scissors in the preparation of generation two samples for the final round of testing.

5.3.3. Overall results of imaging

The images taken at 40x magnification showed significant deformations for both generation one and two samples. This will potentially lead to the results for specific strength being significantly lower than is accurate. There are though greater deformations seen in the

generation two images prepared with scissors than the generation one images prepared using the knife. This will mean the preparation techniques are likely to have caused a greater reduction in the values of strength for generation two samples.

It is also significant that fibres can be seen in the generation two 10x magnification images. These are not seen in the generation one images. Without the presence of a scale on the image the exact length of the fibres cannot be known but at a 10x magnification fibres of the size expected in nano-paper would not be expected to be visible. This is significant as the strength of nano-paper relies on the fibres being small, and so subsequently having a large surface area for bonding to occur. This may lead to the generation two samples not being as strong as the generation one samples. It also shows that the different preparation methods, enzyme or chemical treatment, have a significant effect on sample structure.

5.4. Conclusion

The table and graphs in this section show a summary of the results for both the generation one and two samples.

	Strength / MN/m ²			Specific Strength / KNm/kg		
	Maximum	Average	Standard Deviation	Maximum	Average	Standard Deviation
Generation 1	272.9	173.9	57.5	170.2	103.1	35.3
Generation 2	311.3	108.3	57.7	189.4	68.5	34.3

Figure 23; Table summarising the results of testing for both generation one and two samples

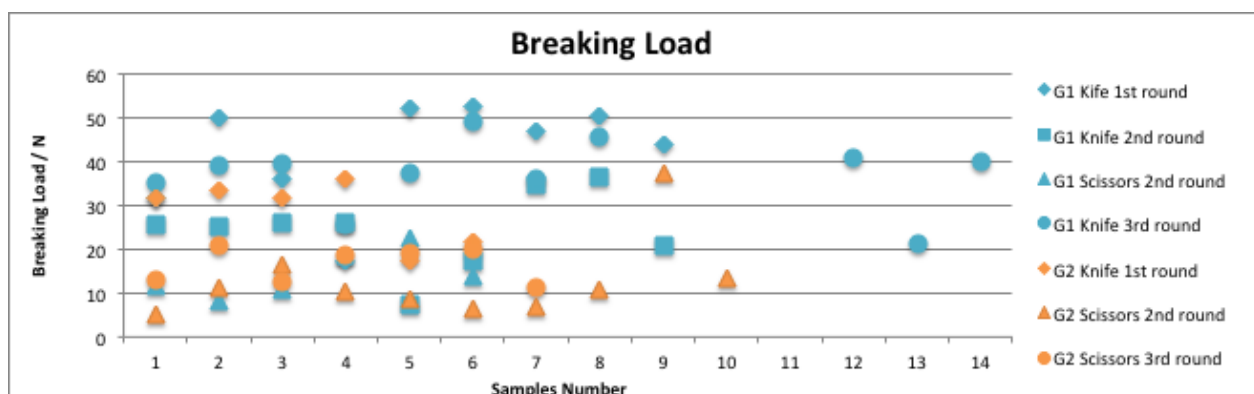


Figure 24; Graph showing the load at which failure occurred for both generation one and two samples

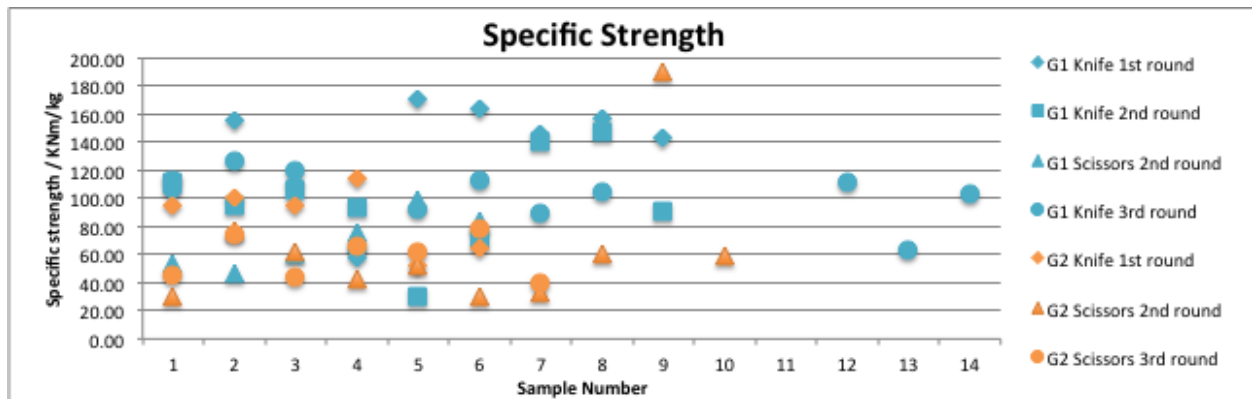


Figure 25; Graph showing the calculated specific strength for both generation one and two samples

The results in section 5 were used to inform the choice of cutting tool used for the generation one and two samples. For the generation one sample a Stanley knife was chosen. This was due to a mixture of the numerical results exceeding those for the scissors, and the images indicating less edge deformation. For the generation two samples there was less difference between the numerical result, but the results of the imaging showed that the scissors were the better choice.

All of the average results were lower than expected for the specific strength, based on the data from Marielle Henriksson (2008). This is likely to be in part due to deformation created during the cutting process, as the images for the chosen tools still showed irregularities. Further the samples could have been contaminated during preparation or testing. If the samples are handled then they can deteriorate. This could be the reason the results for the testing done without card windows, rounds two and three, were lower than for the first round. The lack of the card window made it significantly harder to not handle the samples. Further the lack of a window made it harder to line the sample up correctly, as mentioned in section 4.4.2. This could lead to stress concentrations, and so lower than expected results for the later readings.

Due to the issues with the testing procedure, and the spread of the results, it is difficult to accurately compare the two generations of nano-paper. The results do tend to suggest though that generation one, the enzyme treated one, has the higher specific strength. This is backed

by the results in the table that show the averages to be separated by more than one standard deviation. Also, the graphs show that generation one results shown by the blue dots, tend to be higher than the generation two results, shown by the orange dots.

This difference could be due to the difference in structure of the generation two nano-paper, compared to that of generation one. In the 10x magnification images for the generation two samples it is possible to identify fibres. This is not possible in the generation one images. As the strength of nano-paper is due to the small size of the fibres leading to a large surface area, and so a high number of bonds, large fibres could be a cause of weakness. This could mean that there is a significantly lower bond density in the generation two samples, compared with the generation one samples. This is a possible cause of the lower significant strength and could mean that an enzyme pre-treatment is more efficient for this application.

6. Conclusion

This report has examined two types of cellulose nano-paper; generation one, prepared using enzymes, and generation two, prepared using a chemical pre-treatment. These were tested using a tensile testing machine to calculate their specific strength, and images were taken to assess the preparation procedure. The results are significant in that they provide information on; the mechanical properties, specifically specific strength, the effect of the two different pre-treatments, and the issues with the testing procedure. These issues are of course interlinked and so the conclusion will look at the topic holistically.

6.1. *Testing procedure*

The large spread of results shows that there are issues with the testing procedure. These are likely to occur at both the preparation and testing stages. At the preparation stage was been concluded that there were issues with how the samples were cut. Initially both samples were cut using a Stanley knife but, after experimentation, biological scissors were used for generation two. This was based on the findings of the microscope images which showed large edge defects when a Stanley knife was used to cut generation 2 samples. While the scissors

showed improvements at this level, there was not a significant improvement in the numerical values. Therefore, if further testing was to be carried out, investigation into cutting instruments would be necessary. There may also be the possibility to smooth the edges, getting rid of large defects, by using a fine abrasive material.

At the testing stage, several issues were observed. The first issue observed was with the adhesive used during the first round of testing. Adhesive was used to fix the samples to card windows, which were used to ensure correct alignment of the samples. The samples were observed to have consistently broken at the edge of the area of contact with the card windows. It was concluded that this was likely to have been due to the effect of the glue on the paper, causing chemical degradation of the paper at the point of contact. If this was occurring, it would explain why the samples consistently broke at the point where the glue was applied. This would suggest that the readings for these samples were lower than the true specific strength of the paper. To avoid this impact on the results, rounds 2 and 3 were conducted without gluing the samples to card windows.

As no card window was used for the later stages, there were difficulties correctly aligning the samples, leading to stress concentrations. The grips may have also caused stress concentration and small defects. These could cause lower values of breaking load to be measured. These issues would need to be resolved before further testing was carried out. The possibility of using a different adhesive, that did not affect the samples, is a possibility to solve the alignment issue. Alternatively, different clamps could be used that had a larger surface area, though these could lead to slipping, so this would be more complex to solve.

6.2. *Comparing the two generations of nano-paper*

The values for specific strength lead to the conclusion that the generation one samples were the strongest. This is also backed by the microscope images, which show much larger fibres for generation two. Larger fibres would be expected to cause a reduction in strength, due to a lower surface area per unit of volume for bonding. This is not conclusive though, due to the issues with sample preparation.

The images show a greater amount of edge defects for the generation two samples, even with the use of biological scissors. This is likely to have meant that the samples would have broken at a lower force than they would otherwise have done. Further, the highest value for specific strength is found under generation two. Whilst this is only one result, it does mean that these higher results were possible. Therefore, further testing with a greater number of samples, and an improved testing process is needed for conclusive results to be drawn. Limitations in the availability of samples prevented further rounds of tests from being carried out. However, the tests carried out will have been useful in identifying issues to help to refine the process for further testing.

7. Project management, consideration of sustainability and health and safety

7.1. Project management

The final Gantt chart for this project is included in figure 26 on the next two pages. This summarises the tasks that were completed during the project and how the project was structured. This was a working document and so changed during the course of the project.

The greatest influence on the management of the project was the availability of the tensile testing machine. As the machine is used by the whole department it was crucial to book time and then slot other tasks, such as background reading, in around this. It was also necessary for enough time to be left between each round of testing to allow for results to be calculated and then reviewed. This allowed the testing process to be improved between each round, making later rounds more useful. To ensure testing was as productive as possible, practice tests were carried out during term one, using old samples, so techniques could be practised.

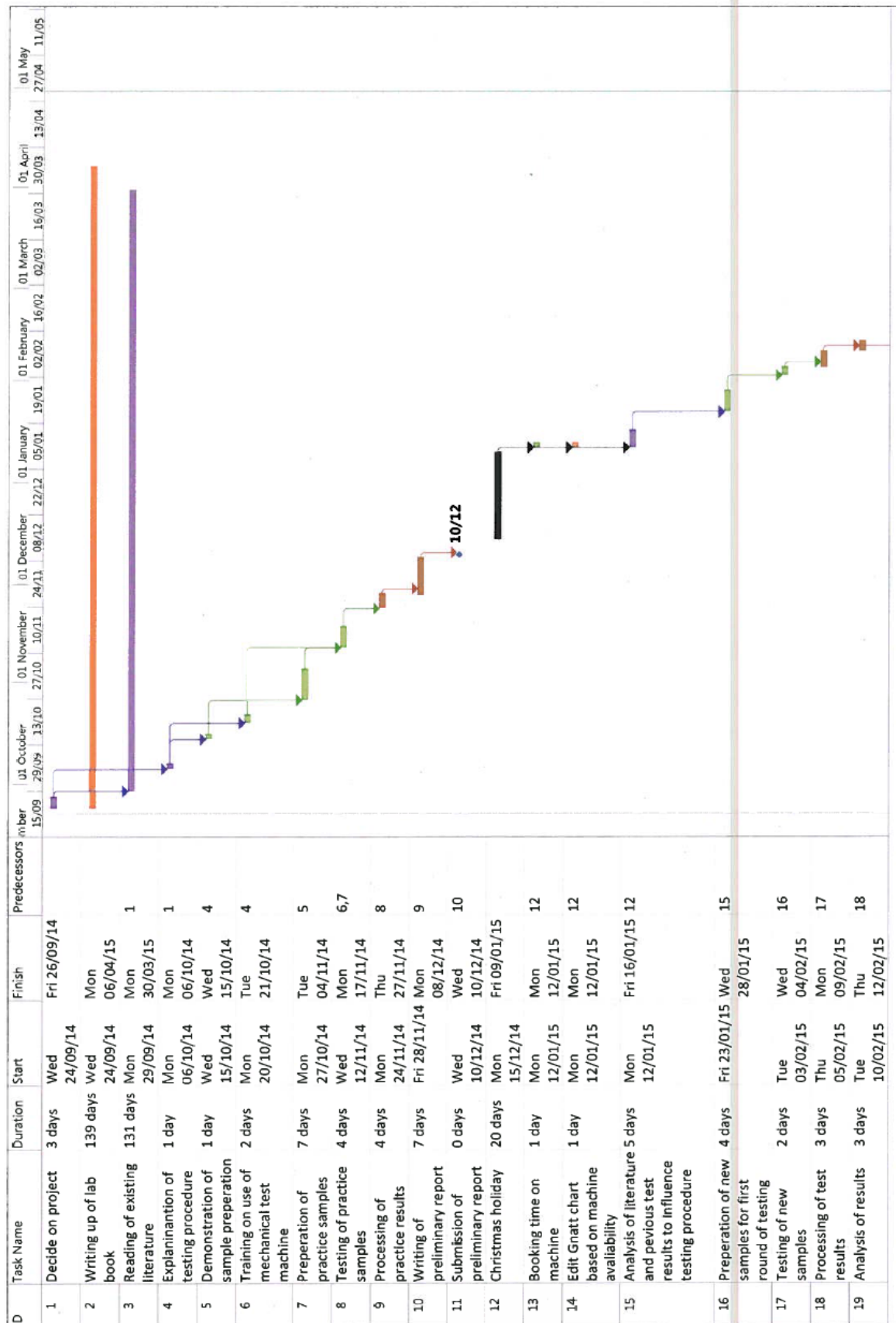


Figure 26; Gantt chart showing break down of project into individual tasks, and when they were completed

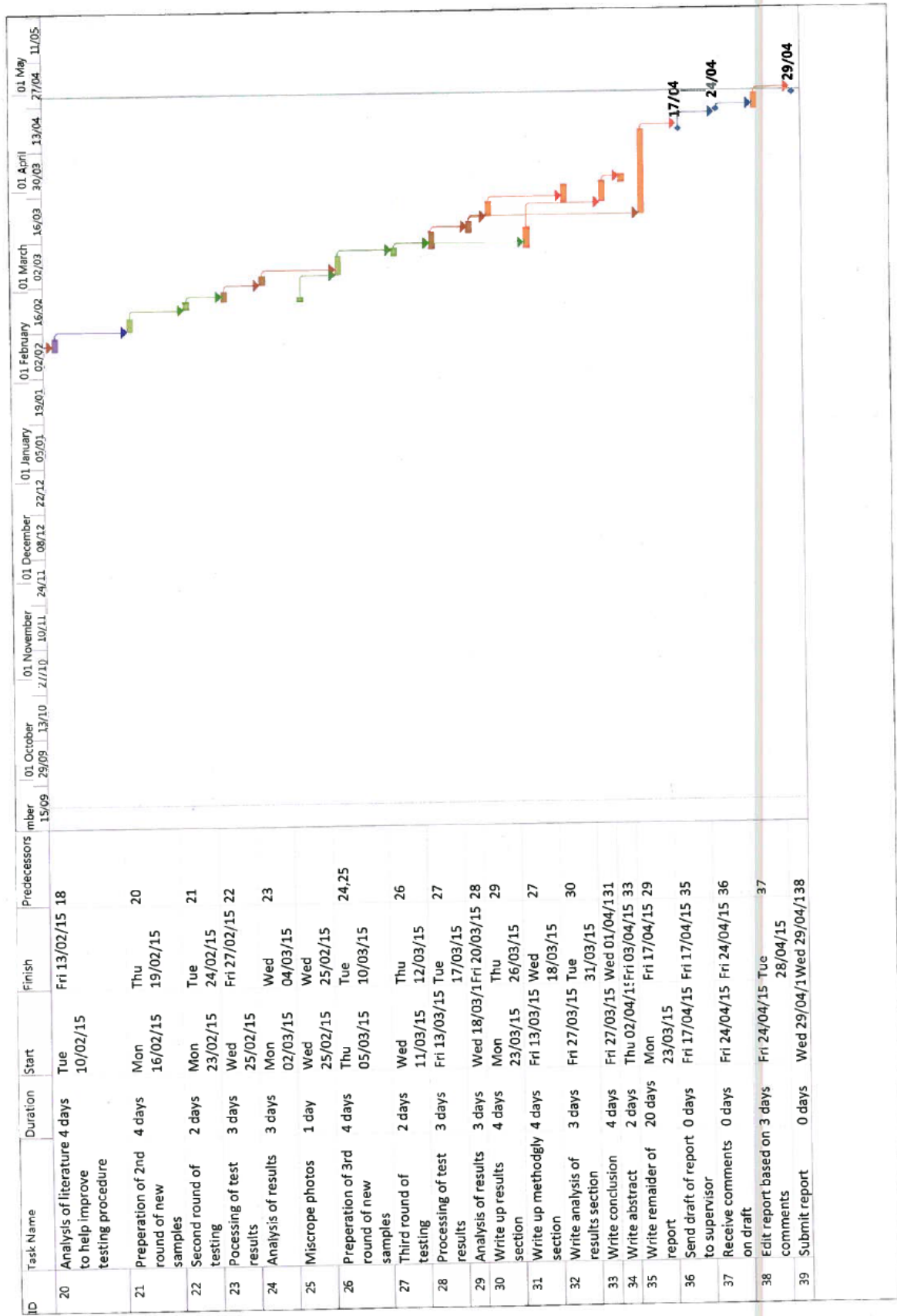


Figure 26; Gantt chart showing break down of project into individual tasks, and when they were completed

7.2. *Health and safety*

The health and safety risks during the project were fairly limited. The project can be separated into three distinct sections; sample preparation, tensile testing, and sample imaging. During sample preparation a Stanley knife was used to cut the samples. It was important therefore to ensure that this was safely used which meant ensuring that cutting was always away from fingers. This meant that, even if slips were to occur, injury would not. Further slips were reduced by the use of a mat on which to cut the samples. As the blade automatically retracted there was no issues with it being left lying around unsheathed. An adhesive was used for the first round of preparation. It was important that this was carried out in a well ventilated area.

There are very limited risks associated with using the tensile testing machine for sample testing, aside from those associated with normal laboratory work. This meant it was necessary to wear appropriate clothing and footwear. The lab area needed to be kept tidy to help reduce trip hazards, and surfaces needed to be kept clear. Further no food or drink was permitted within the testing area and it was important to be aware of other personnel working in the labs. Before testing began, a full demonstration was given by Dr Tommy Shyng, who then observed the initial rounds. This ensured that the machine was used safely and so limited risks. It was necessary to ensure that moving parts of the machine were not touched during operation. Special care needed to be taken during the securing of the samples in the clamps to keep fingers clear. Further, as heavy load cells were moved during the set up and disassembly of the machine, the wearing of sensible, closed-toe shoes was very important.

As sample imaging took place under the supervision of Dr Nandula Wanasekara the health and safety risks were fairly low. The main issue was the use of different types of harmful radiation by machines within the lab. Therefore it was important to ensure the radiation levels measured by the meters around the room were within the safe zone.

Overall the health and safety risks during this project were fairly low. The main risk came from the use of the Stanley knife, but the chance of any serious injury was very low and steps were taken to mitigate these risks. By following good practice, and gaining instruction from

trained personnel, the chances of serious injury were largely negated. Therefore it was felt that the risks were low enough for the project to be carried out.

7.3. *Sustainability*

This project has important sustainability issues. As nano-paper is made from a renewable, recyclable and ecologically “friendly” material it has a big importance ecologically. It is made from cellulose, which is found in plant cells, and so is highly renewable. It could potentially be used instead of polymers to reinforce materials, which would reduce the use of crude oil, a non renewable resource.

This project compares two different types of nano-paper. Generation one is produced using an enzyme pre-treatment, whereas generation two was pre-treated chemically, in a higher energy process. This means that if it can be shown that the generation one samples are stronger, as is shown in this report, then the lower energy method could be used. This would mean a more environmentally friendly production process, meaning the nano-paper would be a more sustainable product. This should also help to reduce costs as energy prices rise.

At the moment the production process is fairly low cost. The raw materials, mainly biomass along with enzymes or chemicals, are cheap because trees can be grown cheaply over much of the world. The production process is not energy intensive, and only the cellulose extraction uses energy. This means that this could be used as a viable alternative to many other reinforcing materials.

References

-
- ⁱ S.J.Eichhorn et al. 2010. Review: current international research into cellulose nanofibres and nanocomposites. *Journal of Material Science*, volume 45, issue 1, pp 1-33.
- ⁱⁱ Klemm, D., Kramer, F., Moritz, S., Lindström, T., Ankerfors, M., Gray, D. and Dorris, A. (2011), Nanocelluloses: A New Family of Nature-Based Materials. *Angew. Chem. Int. Ed.*, 50: 5438–5466. doi: 10.1002/anie.201001273
- ⁱⁱⁱ Guangyuan Zheng et al. 2013, Nanostructured paper for flexible energy and electronic devices, *Mrs Bulletin*, Volume 38, available at www.mrs.org/bulletin, accessed 19/04/15
- ^{iv} Marielle Henriksson, Lars A. Berglund, Per Isaksson, Tom Lindström, and Takashi Nishino, 2008, Cellulose Nanopaper Structures of High Toughness, *Biomacromolecules* 9 (6), 1579-1585 DOI: 10.1021/bm800038n
- ^v ASM International, 2004, Tensile testing, second edition, chapter 1. Available from <http://www.asminternational.org/> accessed 20/04/15
- ^{vi} Fracture Mechanics, Chapter 10, Dr.ir. P.J.G. Schreurs, 2012, Available from www.mate.tue.nl/~piet/edu/frm/pdf/frmsyl1213.pdf, accessed 18/04/15
- ^{vii} M.F. Ashby, D.R.H. Jones, 2012, *Engineering Materials 1*, Forth Edition, Butterworth-Heinemann, pg 122.